Exponents

An **exponent** is a shorthand notation for repeated multiplication of a number by itself.

$$a^n = a \cdot a \cdot a \ldots . \cdot a \ (n\ times)$$

where a is the base and n is the exponent.

Examples:

a) $10^3 =$ b) $2^5 =$

Notes:

i) $a^1 =$ ii) $a^0 =$ $(a \neq 0)$

3.1 Describe a decimal number in expanded notation

The decimal system is the most widely used number system. It is a base 10 system and uses digits 0 through 9.

Place Value

From right to left, the places are ones, tens, hundreds, thousands, ten thousands, and so on.

… 10 000 1 000 100 10 1

Each place value can be written as a power of 10:

$$1 = 10^0 \ ; \ 10 = 10^1 \ ; \ 100 = 10^2 \ ; 1\,000 = \ 10^3 \ ; 10\,000 = 10^4 \ ; \ \ldots$$

Moving from right to left, the exponent increases by one at each place.

3.2 Express a decimal number in expanded notation

To write a decimal number in expanded exponential notation, multiply each digit by its corresponding power of 10 and add the resulting terms.

Examples

Express $5\,427$ in expanded exponential form.

$5\,427 = 5 \times 10^3 + 4 \times 10^2 + 2 \times 10^1 + 7 \times 10^0$

Express $36\,891$ in expanded exponential form.

3.3 Describe a binary number in expanded notation

The binary system is a base 2 number system. The only digits used are 1 and 0.

A subscript 2 identifies a number as binary, for example 111_2 and 1101_2.

Place Value

From right to left, the places are ones, twos, fours, eights, sixteens, and so on.

… 16 8 4 2 1

Each place value is a power of 2:

$$1 = 2^0 \;;\; 2 = 2^1 \;;\; 4 = 2^2 \;;\; 8 = 2^3 \;;\; 16 = 2^4 \ldots..$$

In exponent form, the place values are

$$\ldots\ldots, 2^4, 2^3, 2^2, 2^1, 2^0.$$

To expand a binary number, start at the rightmost digit (2^0). Moving left, increase the exponent by one for each position. Then write the expansion from left to right.

3.4 Express a binary number in expanded notation

Examples

Expand 1101_2 as a sum of appropriate powers of 2.

- First, observe that the highest exponent place value in this number is 2^3.
- Expanding as with the Decimal System earlier, we get:

$$1\ 1\ 0\ 1_2 = 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 2^3 + 2^2 + 0 + 1$$

Since binary digits are either 0 or 1, we can simplify the expansion by including only the powers of 2 corresponding to digits that are 1.

Evaluating and adding these powers gives the decimal equivalent.

$$2^3 + 2^2 + 0 + 1$$

$$8 + 4 + 1$$

$$13_{10}$$

Binary Place-Value Reference

2^5	2^4	2^3	2^2	2^1	2^0
32	16	8	4	2	1

By first expanding using exponents, convert each binary number to its decimal equivalent.

a) $1\ 0\ 1\ 1\ 0_2$

b) $1\ 1\ 0\ 1\ 0\ 1_2$

c) $1\ 1\ 1\ 1\ 0\ 0\ 1_2$

Computer Applications

Computers use the binary number system to represent and store information. Binary uses only two digits, 0 and 1, which correspond to two possible electronic states.

Each binary digit (0 or 1) is referred to as a bit.

In computers, bits are commonly arranged in groups of eight.

Each group of eight bits is referred to as a byte.

Place value for 8-digit binary numbers

Bit positions are counted from right to left, beginning with the first position on the right.

Bit Location	8th	7th	6th	5th	4th	3rd	2nd	1st
Exponent								
Place Value								

Problems:

1. Identify the place value of the 1 in each byte.

   a) $0\,0\,0\,1\,0\,0\,0\,0_2$

   b) $0\,1\,0\,0\,0\,0\,0\,0_2$

2. Convert each binary representation to its decimal equivalent. Show your expansion.

   a) $0\,1\,0\,1\,0\,1\,1\,1_2$

   b) $1\,1\,0\,0\,1\,1\,0\,1_2$

3.9 Convert whole numbers using the following systems:

3.9.1 Decimal

3.9.2 Binary

Converting Decimal Numbers to Binary

Question: How would we convert a Decimal number to Binary form?

Answer: Determine which powers of 2 add up to the decimal number. Write 1 in the positions corresponding to powers of 2 that are needed, and 0 in the remaining positions.

Note

Every nonnegative integer has a unique binary representation when leading zeros are omitted. For example, 1101_2 and 00001101_2 represent the same value.

Express 13 in binary form.

Solution:

First, find the largest power of 2 less than or equal to 13. Since $2^4 = 16$ and $2^3 = 8$, the largest needed place value is 8.

Place Value	8	4	2	1
Digit	1			

Subtract 8 from 13 to get 5. Since 5 = 4 + 1, place a 1 in the fours and ones positions. The twos position is 0.

Place Value	8	4	2	1
Digit	1	1	0	1

Therefore, $13_{10} = 1101_2$.

Examples

Express each decimal number in binary form. Show your work.

a) 58

b) 118

c) 231

Larger Numbers

Question 1: What is the largest nonnegative decimal number that can be represented by an 8-bit unsigned binary number?

The largest unsigned 8-bit value occurs when all eight digits are 1: 11111111_2.
Expand using powers of 2, then evaluate:

Question 2: How could the decimal number 256 be represented in binary form?
$256 = 2^8$. This requires a ninth bit, so its binary representation has nine digits.

Question 3: What is the largest nonnegative decimal number that can be represented by a 9-bit unsigned binary number?
Begin with 111111111_2, then expand and evaluate.

Look for a pattern in your answers to Questions 1 and 3. The largest unsigned value represented by n bits is:

$$2^n - 1$$

Subsequent place values beyond 256 (2^8) are:

i) 2^9 = _____ (10-digit binary numbers)

ii) 2^{10} = _____ (11-digit binary numbers)

Examples

1. Convert the following binary numbers to decimal form.

a) $1\quad 0\quad 1\quad 1\quad 0\quad 0\quad 0\quad 0\quad 1\quad 1_2$

b) $1\quad 1\quad 0\quad 1\quad 0\quad 1\quad 0\quad 1\quad 0\quad 0\quad 1_2$

2. Express 375 in binary form.

3.5 Describe an octal number in expanded notation

The octal number system is a base 8 number system. It uses eight digits: 0, 1, 2, 3, 4, 5, 6, and 7.

A subscript 8 identifies a number as octal. For example, $2\ 5\ 3_8$ and 1372_8.

Like the decimal and binary number systems, the octal system uses place values. However, each place value is a power of 8 rather than a power of 10 or 2.

3.6 Express an octal number in expanded notation

Place Value

From right to left, the place values are ones, eights, sixty-fours, five hundred twelves, and so on.

… , 512 , 64 , 8 , 1

Each place value is a power of 8:

Power of 8	Place Value
8^0	1
8^1	8
8^2	64
8^3	512
8^4	4096

In exponent form, the place values are

$$\ldots 8^4, 8^3, 8^2,\ 8^1,\ 8^0$$

Example

Express 253_8 in expanded notation.

3.9 Convert whole numbers using the following systems:

3.9.1 Decimal

3.9.2 Binary

3.9.3 Octal

Octal to Decimal Conversions

To convert an octal number to decimal:

- Multiply each digit by its corresponding power of 8.
- Add the results.

Convert 253_8 to decimal.

Convert 1372_8 to decimal.

Converting Decimal to Octal

To convert a decimal number to octal:

- Divide the decimal number by 8.
- Record the remainder.
- Continue dividing the quotient by 8 until the quotient is 0.
- Read the remainders from bottom to top to obtain the octal number.

Convert 156_{10} to octal.

Convert 231_{10} to octal form.

Convert the following decimal numbers to octal form.

a) 311

b) 699

c) 1875

Binary to Octal Conversions

To convert a binary number to octal, we can first convert it to decimal and then convert the decimal number to octal. However, there is a much easier method.

- Starting from the right, divide the binary number into groups of three digits (bits).
- If the leftmost group has fewer than three bits, add leading zeros.
- Convert each group of three bits to its decimal equivalent (0 through 7).
- Write the resulting digits in the same order to obtain the octal number.

Example

Convert 1101011_2 to octal.

Solution:

Group the digits from left to right.

001 101 011

Convert each group.

$$001 = 2^0 = 1$$
$$101 = 2^2 + 2^0 = 4 + 1 = 5$$
$$011 = 2^1 + 2^0 = 2 + 1 = 3$$

Therefore, $1101011_2 = 153_8$.

Convert the following binary numbers to octal form.

110010001_2

1010011111_2

11101011001_2

Converting Octal to Binary

To convert an octal number to binary, we reverse the process.

- Convert each octal digit to its three-bit binary equivalent.
- Combine the groups in the same order.
- Remove any unnecessary leading zeros.

Example

Convert 347_8 to binary.

Solution:

Convert each digit.

$$3 = 011$$
$$4 = 100$$
$$7 = 111$$

Combine the groups.

$$011\ 100\ 111$$

Remove the unnecessary leading zero.

$$347_8 = 11100111_2$$

Convert the following octal numbers to binary form.

621_8

4237_8

3052_8

3.7 Describe a hexadecimal number in expanded notation

The hexadecimal number system is a base 16 number system.

It uses sixteen symbols:

- The digits 0-9
- The letters A–F, representing the values 10-15.

Hexadecimal	Decimal Value
A	10
B	11
C	12
D	13
E	14
F	15

A subscript 16 identifies a number as a hexadecimal.

For example, 256_{16} and $7A4_{16}$.

3.8 Express a hexadecimal number in expanded notation

Each position in a hexadecimal number represents a power of 16.

Power of 16	Place Value
16^0	1
16^1	16
16^2	256
16^3	4096

Expanded notation means writing each digit multiplied by its place value. For a letter, substitute its decimal value.

Example

Write $7A4_{16}$ in expanded notation.

$7A4_{16} = 7 \times 16^2 + 10 \times 16^1 + 4 \times 16^0$

Write it as its decimal equivalent.

$7A4_{16} = 7 \times 16^2 + 10 \times 16^1 + 4 \times 16^0 = 7 \times 256 + 10 \times 16 + 4 \times 1 = 1792 + 160 + 4$
$7A4_{16} = 1956_{10}$

Comparison of four number systems

Decimal (base 10)	Binary (base 2)	Octal (base 8)	Hexadecimal (base 16)
0	0_2	0_8	0_{16}
1	1_2	1_8	1_{16}
2	10_2	2_8	2_{16}
3	11_2	3_8	3_{16}
4	100_2	4_8	4_{16}
5	101_2	5_8	5_{16}
6	110_2	6_8	6_{16}
7	111_2	7_8	7_{16}
8	1000_2	10_8	8_{16}
9	1001_2	11_8	9_{16}
10	1010_2	12_8	A_{16}
11	1011_2	13_8	B_{16}
12	1100_2	14_8	C_{16}
13	1101_2	15_8	D_{16}
14	1110_2	16_8	E_{16}
15	1111_2	17_8	F_{16}
16	10000_2	20_8	10_{16}

3.9 Convert whole numbers using the following systems:

3.9.1 Decimal
3.9.2 Binary
3.9.3 Octal
3.9.4 Hexadecimal

Hexadecimal to Decimal Conversions

- Multiply each hexadecimal digit by its corresponding power of 16.
- Add the results.

Convert 256_{16} to a decimal.

Convert 1839_{16} to a decimal.

Convert $7A4_{16}$ to a decimal.

Convert $B02C_{16}$ to a decimal.

Decimal to Hexadecimal Conversion

- Divide by 16.
- Record the whole-number quotient and remainder (0 - 15).
- Convert remainders 10 - 15 into A - F.
- Repeat using each quotient until it reaches 0.
- Read the remainders upward.
    - The final quotient is NOT an additional digit.

Convert 598_{10} to its hexadecimal.

Convert 1956_{10} to its hexadecimal.

Convert 3608_{10} to its hexadecimal.

Convert 5997_{10} to its hexadecimal.

Binary to Hexadecimal Conversions

Since 16 = 2^4, each hexadecimal digit corresponds to FOUR binary bits.

- Group bits into fours from the right, adding leading zeros on the left if necessary.
- Convert each 4-bit representation to its total decimal value and the resulting digits/letters form the required hexadecimal number.

Example

What is the hexadecimal equivalent of the binary number 10100110_2?

Solution:

Starting from the right, group the binary digits into sets of four.

1010 0110

Convert each group:

$$1010 = 2^3 + 2^1 = 8 + 2 = 10 \rightarrow A$$
$$0110 = 2^2 + 2^1 = 4 + 2 = 6$$

Therefore, $10100110_2 = A6_{16}$

Convert 11001111_2 to hexadecimal.

Convert 1010011101_2 to hexadecimal.

Convert 111100111_2 to hexadecimal.

Hexadecimal to Binary

We simply reverse the above process.

Replace each hexadecimal digit with its four-bit binary equivalent. Keep four bits per digit while converting; leading zeros in the final number may then be omitted.

Example

Convert $A4C_{16}$ to binary.

Solution:

$$A = 10_{10} = 1010_2$$

$$4 = 0100_2$$

$$C = 12 = 1100_2$$

Therefore, $A4C_{16} = 101001001100_2$.

Convert $F38_{16}$ to binary.

Convert $1B29_{16}$ to binary.

Convert $A5D4_{16}$ to binary.

Octal and Hexadecimal Conversions

There is no direct fixed-size digit grouping between octal and hexadecimal.

- Convert octal to binary (3 bits per digit).
- Regroup into fours for hexadecimal.

Reverse the steps for hexadecimal to octal. You can also convert through decimal.

Example

Convert 75_8 to hexadecimal.

Solution:

Convert each octal digit to its 3-bit binary equivalent.

Octal digit	7	5
Binary equivalent	111	101

Therefore, $75_8 = 111101_2$.

Starting from the right, group the binary digits into sets of four. Add leading zeros if necessary.

0011 1101

Convert each group to its hexadecimal equivalent.

Binary group	0011	1101
Hexadecimal digit	$2^1 + 2^0$ $2 + 1$ 3	$2^3 + 2^2 + 2^0$ $8 + 4 + 1$ $13 \rightarrow D$

Therefore, $75_8 = 3D_{16}$.

Convert 573_8 to its hexadecimal equivalent.

Convert $2D7_{16}$ to its octal equivalent.

Number System Conversions | Quick Reference

Decimal (base 10) • Binary (base 2) • Octal (base 8) • Hexadecimal (base 16)

Conversion	Method
Decimal → Binary	Divide repeatedly by 2. Read remainders from bottom to top.
Binary → Decimal	Multiply each digit by its corresponding power of 2. Add.
Decimal → Octal	Divide repeatedly by 8. Read remainders from bottom to top.
Octal → Decimal	Multiply each digit by its corresponding power of 8. Add.
Decimal → Hexadecimal	Divide repeatedly by 16. Read remainders bottom to top; use A - F for 10 - 15.
Hexadecimal → Decimal	Multiply each digit by its corresponding power of 16. Add (A = 10, …, F = 15).
Binary → Octal	Group bits in sets of 3 from the right. Convert each group to an octal digit.
Octal → Binary	Replace each octal digit with its 3-bit binary equivalent.
Binary → Hexadecimal	Group bits in sets of 4 from the right. Convert each group to a hexadecimal digit.
Hexadecimal → Binary	Replace each hexadecimal digit with its 4-bit binary equivalent.
Octal → Hexadecimal	Convert octal to binary. Regroup bits in sets of 4 from the right.
Hexadecimal → Octal	Convert hexadecimal to binary. Regroup bits in sets of 3 from the right.

Mixed Practice

1. Convert $3E7_{16}$ to decimal.

2. Convert 2745_{10} to hexadecimal.

3. Convert 10111101011_2 to hexadecimal.

4. Convert $D06_{16}$ to binary.

5. Convert 110101101_2 to octal.

6. Convert 647_8 to binary.

7. Convert 734_8 to hexadecimal through binary.

8. Convert $9B_{16}$ to octal through binary.

9. Explain why $2F_8$ is not valid.

10. Check 256_{16} by converting your decimal answer back to hexadecimal.

3.10 Use binary numbers to do the following:

3.10.1 Add

3.10.2 Subtract

3.10.3 Multiply

Addition

Binary addition works much like decimal addition. We add the digits in each column, starting from the right and moving left. However, because binary uses only 0 and 1, we must carry whenever the sum in a column is 2 or greater.

Rules for Binary Addition

Addition	Result	Explanation
0 + 0	0	Write 0
0 + 1	1	Write 1
1 + 0	1	Write 1
1 + 1	10	Write 0, carry 1
1 + 1 + 1	11	Write 1, carry 1

Remember: $10_2 = 2_{10}$. When adding 1+1 in binary, the answer is 10_2, not 2.

Note

We can check our answers by converting the binary numbers to decimal.

Examples

Add $1010_2 + 0101_2$

Add $1011_2 + 0110_2$

Add $101_2 + 110_2 + 011_2$

Add $1010_2 + 0011_2$

Add $1110_2 + 0111_2$

Add 101101_2 and 111011_2. Check your answer by converting all three binary numbers to decimal.

A computer adds two unsigned 8-bit binary numbers: 11111111_2 + 00000001_2.

What is the result? How many bits are needed to represent the answer?

Subtraction

Binary subtraction works much like decimal subtraction. We subtract the digits in each column, starting from the right and moving left. Since binary is a base 2 number system, it uses only two digits: 0 and 1. When the top digit is smaller than the bottom digit, we must borrow from the column to the left.

Rules for Binary Subtraction

Subtraction	Result	Explanation
0 – 0	0	Write 0
1 – 0	1	Write 1
1 – 1	0	Write 0
0 – 1	1	Borrow 1 from the next column to the left

Note

When we borrow 1 from the next column, it is worth 10_2 (or 2 in decimal) in the current column.

Therefore, $10_2 - 1_2 = 1_2$.

Examples

Subtract $1010_2 - 0010_2$.

Subtract $1101_2 - 0110_2$.

Borrowing Across Zeros

Sometimes the digit immediately to the left is 0, so we cannot borrow directly from that column. In this situation, we must move farther to the left until we find a 1. We then borrow from that 1 and work back through the zeros.

- Find the nearest 1 to the left.
- Change that 1 to 0.
- Each zero passed through becomes 1 after passing a borrow to the next column.
- The column where borrowing was needed receives 10_2.

Subtract $10000_2 - 00001_2$.

- Start in the rightmost column.
    - We need to subtract 0 - 1, so we must borrow.
- Look at the column immediately to the left.
    - It contains 0, so we cannot borrow directly from it.
    - Continue moving left until we reach the 1 in the leftmost column.
- Borrow from the leftmost 1.
    - The leftmost 1 becomes 0. The borrowed value passes through each zero until it reaches the rightmost column.
    - The three intermediate zeros each become 1 after passing along the borrow.
    - The rightmost column now contains 10_2.
- Complete the subtraction.
    - Rightmost column: $10_2 – 1 = 1$
    - Next three columns: $1 – 0 = 1$
    - Leftmost column: $0 – 0 = 0$

Therefore, $10000_2 - 00001_2 = 01111_2$

Subtract 1110_2 - 0101_2.

Subtract 11010_2 - 01011_2.

Subtract 100000_2 - 001101_2.

Multiplication

Binary multiplication works much like decimal multiplication. We multiply each digit of the second number by the first number, starting from the right.

Since binary uses only the digits 0 and 1, multiplication is relatively straightforward.

Rules for Binary Multiplication

Multiplication	Result
0 × 0	0
0 × 1	0
1 × 0	0
1 × 1	1

Note

- Multiplying a binary number by 0 gives 0.
- Multiplying a binary number by 1 gives the original number.
- When multiplying by a digit farther to the left, shift the partial product one additional position to the left.
- Add the partial products using the rules of binary addition.

Examples

Multiply 1011_2 x 1_2.

Multiply 101_2 x 11_2.

Multiply 1101_2 x 101_2.

Multiply 1101_2 x 1011_2. Check your answer by converting the numbers to decimal.

Explain why multiplying a binary number by 10_2 is equivalent to shifting all its digits one position to the left.

3.11 Using hexadecimal numbers:

3.11.1 Add

3.11.2 Subtract

3.11.3 Multiply

Addition

To add hexadecimal numbers,

- Line up the digits by place value
- Add from right to left
- Convert to decimal and add the numbers
    - If a column totals 16 or more, subtract 16, write the remainder as a hex digit, and carry 1 to the next column.
    - If the column totals less than 16, convert the number to a hexadecimal

Examples

$2A_{16} + 1F_{16}$

Solution:

Ones column: A + F = 10 + 15 = 25. Since 25 = 16 + 9, write 9 and carry 1.

Sixteens column: 2 + 1 + 1 (carry) = 4.

Answer: 49_{16}. Check: 42 + 31 = 73, and $49_{16} = 73_{10}$.

$7D_{16} + 8B_{16}$

$3B_{16} + 24_{16}$

$6F_{16} + 2D_{16}$

$217_{16} + 1A5_{16}$

Subtraction

To subtract hexadecimal numbers,

- Line up the digits by place value
- Subtract from right to left
    - If the top digit is smaller, borrow 1 from the next place to the left.
    - Borrowing 1 adds 16 to the current column, while reducing the next column by 1.

Examples

$5C_{16} - 2E_{16}$

Solution:

Ones: C (12) is smaller than E (14). Borrow 1 from the 5, so the ones column becomes 12 + 16 = 28.

Ones: 28 − 14 = 14 = E. Sixteens: 4 − 2 = 2.

Answer: $2E_{16}$. Check: 92 − 46 = 46.

$100_{16} - 1A_{16}$

$8D_{16} - 3A_{16}$

$200_{16} - B7_{16}$

$FC4_{16} - 2C7_{16}$

Multiplication

To multiply hexadecimal numbers,

- Multiply the digits as you would in decimal form.
- Convert each product into a hexadecimal.
    - Divide by 16 to find the carry and the remainder.
- For multi-digit multiplication, shift each partial product one hexadecimal place left as needed.
- Add the partial products in base 16.

Examples

$B_{16} \times 6_{16}$

Solution:

B = 11, so 11 × 6 = 66.

66 = 4 × 16 + 2, so the result is 42_{16}.

$2A_{16} \times 3_{16}$

$1A_{16} \times 12_{16}$

$C_{16} \times 4_{16}$

$3D_{16} \times 2_{16}$

$1B_{16} \times 13_{16}$

$A01_{16} \times 29_{16}$

x	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	10
1	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	10
2	2	4	6	8	A	C	E	10	12	14	16	18	1A	1C	1E	20
3	3	6	9	C	F	12	15	18	1B	1E	21	24	27	2A	2D	30
4	4	8	C	10	14	18	1C	20	24	28	2C	30	34	38	3C	40
5	5	A	F	14	19	1E	23	28	2D	32	37	3C	41	46	4B	50
6	6	C	12	18	1E	24	2A	30	36	3C	42	48	4E	54	5A	60
7	7	E	15	1C	23	2A	31	38	3F	46	4D	54	5B	62	69	70
8	8	10	18	20	28	30	38	40	48	50	58	60	68	70	78	80
9	9	12	1B	24	2D	36	3F	48	51	5A	63	6C	75	7E	87	90
A	A	14	1E	28	32	3C	46	50	5A	64	6E	78	82	8C	96	A0
B	B	16	21	2C	37	42	4D	58	63	6E	79	84	8F	9A	A5	B0
C	C	18	24	30	3C	48	54	60	6C	78	84	90	9C	A8	B4	C0
D	D	1A	27	34	41	4E	5B	68	75	82	8F	9C	A9	B6	C3	D0
E	E	1C	2A	38	46	54	62	70	7E	8C	9A	A8	B6	C4	D2	E0
F	F	1E	2D	3C	4B	5A	69	78	87	96	A5	B4	C3	D2	E1	F0
10	10	20	30	40	50	60	70	80	90	A0	B0	C0	D0	E0	F0	100

3.12 Use the AND operator to compare two binary values

The AND operator is used to compare two binary values, one bit at a time.

The result is 1 only when both bits being compared are 1. Otherwise, the result is 0.

The AND operator is commonly used in computer programming and digital electronics. It can be used to check whether specific bits are set to 1 and to perform operations called bit masking.

Rules for the AND Operator

First Bit	Second Bit	Result
0	0	0
0	1	0
1	0	0
1	1	1

Steps for Using the AND Operator

- Write the two binary numbers one above the other, aligning the digits by place value. Add leading zeros if necessary.
- Compare the bits in each column.
- Write 1 if both bits are 1. Otherwise, write 0.
- Continue until all columns have been compared.

Note: The AND operator does not add or subtract binary numbers. There is no carrying or borrowing.

Examples

Use the AND operator to compare 1101_2 and 1011_2.

Use the AND operator to compare 10110_2 and 01101_2

Show all steps for each conversion. Clearly identify the number system of each answer using the appropriate base (subscript).

1. Convert each of the following binary numbers to its octal, decimal, and hexadecimal equivalents.

111011101_2

10101010111_2

111100000_2

2. Convert each of the following octal numbers to its binary, decimal, and hexadecimal equivalents.

3754_8

7777_8

247_8

3. Convert each of the following decimal numbers to its binary, octal, and hexadecimal equivalents.

3479_{10}

642_{10}

555_{10}

4. Convert each of the following hexadecimal numbers to its binary, octal, and decimal equivalents.

$4FB2_{16}$

$88BAE_{16}$

$DC4_{16}$

MA1905 Operations with Binary and Hexadecimal Numbers

Show all carries, borrowing, and partial products. Give answers in the same number base as the question. Do not convert to decimal as your main method.

Add the following binary numbers.

1.

$$\begin{array}{r} 1011 \\ +\ 110 \\ \hline \end{array}$$

2.

$$\begin{array}{r} 1101 \\ +\ 1011 \\ \hline \end{array}$$

3.

$$\begin{array}{r} 10101 \\ +\ 1110 \\ \hline \end{array}$$

4.

$$\begin{array}{r} 11101 \\ +\ 10111 \\ \hline \end{array}$$

5.

$$\begin{array}{r} 101101 \\ +\ 11011 \\ \hline \end{array}$$

6.

$$\begin{array}{r} 111011 \\ +\ 101101 \\ \hline \end{array}$$

Subtract the following binary numbers.

7.

$$\begin{array}{r} 1101 \\ -\ 101 \\ \hline \end{array}$$

8.

$$\begin{array}{r} 10110 \\ -\ 1001 \\ \hline \end{array}$$

9.

$$\begin{array}{r} 11001 \\ -\ 1011 \\ \hline \end{array}$$

10.

$$\begin{array}{r} 100000 \\ -\ 1101 \\ \hline \end{array}$$

11.

$$\begin{array}{r} 101101 \\ -\ 1110 \\ \hline \end{array}$$

12.

$$\begin{array}{r} 110010 \\ -\ 10111 \\ \hline \end{array}$$

Multiply the following binary numbers. Show all partial products.

13.

$$\begin{array}{r} 101 \\ \times\ 11 \\ \hline \end{array}$$

14.

$$\begin{array}{r} 110 \\ \times\ 101 \\ \hline \end{array}$$

15.

$$\begin{array}{r} 1011 \\ \times\ 11 \\ \hline \end{array}$$

16.

$$\begin{array}{r} 1101 \\ \times\ 101 \\ \hline \end{array}$$

17.

$$\begin{array}{r} 1010 \\ \times\ 110 \\ \hline \end{array}$$

18.

$$\begin{array}{r} 1110 \\ \times\ 1011 \\ \hline \end{array}$$

Add the following hexadecimal numbers.

19.

$$\begin{array}{r} 2A \\ +\ 17 \\ \hline \end{array}$$

20.

$$\begin{array}{r} 3F \\ +\ 28 \\ \hline \end{array}$$

21.

$$\begin{array}{r} 7B \\ +\ 4D \\ \hline \end{array}$$

22.

$$\begin{array}{r} A6 \\ +\ 39 \\ \hline \end{array}$$

23.

$$\begin{array}{r} 1E8 \\ +\ 2B7 \\ \hline \end{array}$$

24.

$$\begin{array}{r} 9AF \\ +\ 6D5 \\ \hline \end{array}$$

Subtract the following hexadecimal numbers.

25.

$$\begin{array}{r} 5C \\ -\ 27 \\ \hline \end{array}$$

26.

$$\begin{array}{r} A4 \\ -\ 3B \\ \hline \end{array}$$

27.

$$\begin{array}{r} F2 \\ -\ 6D \\ \hline \end{array}$$

28.

$$\begin{array}{r} 10A \\ -\ 7F \\ \hline \end{array}$$

29.

$$\begin{array}{r} 3B5 \\ -\ 1C8 \\ \hline \end{array}$$

30.

$$\begin{array}{r} A20 \\ -\ 4E9 \\ \hline \end{array}$$